

# How much strength is necessary?

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The purpose of this review was to consider the association of measures of maximum strength in relation to sports performance and performance variables, which rely on high levels of power and speed, in essence it is an expansion of the ideas and concepts presented by Stone and Moir (2001). Evidence from different types of cross-sectional research as well as observational data was considered. Collectively the data indicate that the association between maximum strength and sport performance related variables such as peak power and peak rate of force development is quite strong. While explaining performance in strength/power sports is a multi-factorial problem, there is little doubt that maximum strength is a key component. © 2002 Published by Elsevier Science Ltd.

## Introduction

Strength can be defined as the ability to produce force (Siff 2001, Stone 1993). Thus, the display of strength would have characteristics including a magnitude (0–100%), a rate and a direction. Furthermore the force generation may be isometric or dynamic. The level of force production and its characteristics are determined by a number of factors including the type of contraction, the rate of muscle activation and the degree of muscle activation. The importance of force production can be ascertained from Newton's second law:

$$F = ma$$

Thus acceleration ( $a$ ) of a mass ( $m$ ) such as body mass or an external object depends upon the ability of the musculature to generate force ( $F$ ). Furthermore, power production is the product of force and velocity and is likely the most important factor in determining success in most sports. Thus, the 'ability to generate force (strength)' is an integral part of power production and therefore may be a key component in determining athletic success.

Generally, most coaches and athletes agree that in sports such as weightlifting and particularly powerlifting, continuous increases in maximum strength would be advantageous. However, there is no agreement with regard to how much strength is necessary for most sports.

The purpose of this discussion is to describe the *potential* relationship between strength and (1) sports performance and (2) other variables contributing to sports performance, particularly rate of force development and power. This discussion will be confined to those sports generally considered to rely heavily on high levels of strength and power (i.e. 'strength-power' sports).

## Strength/power sports

From the perspective of this discussion, there are two variables of primary importance for most sports (1) the peak rate of force development (PRFD) and (2) power output. The PRFD is associated with the concept of 'explosive strength' and is directly related to the ability to accelerate objects including body mass (Schmidtbleicher 1992).

Work is the product of force and the distance that the object moves in the direction of the force (force  $\times$  distance). Power is the rate of doing work ( $P = \text{force} \times \text{distance}/\text{time}$ ) and can be expressed as the product of force and speed ( $P = \text{force} \times \text{speed}$ ). Power can be calculated as an average over a range of motion or as an instantaneous value occurring at a particular instant during the displacement of an object. Peak power (PP) is the highest instantaneous power value found over a range of motion. Maximum power (MP) is the highest

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peak power output one is capable of generating under a given set of conditions (i.e. state of training, type of exercise, etc.). Muscular actions that maximise power include jumping, throwing and kicking; indeed activities in which a movement sequence results in maximum achievable velocities primarily depends upon power production (Young 1993). Furthermore, activities requiring a rapid direction change and acceleration, such as displays of 'agility', depend upon bursts of high power output. Thus, power output is likely to be the most important factor in separating sports performances (i.e. who wins and who loses). Although average power output may be more associated with performance in endurance events, for activities such as jumping, sprinting and weightlifting movements PP is typically strongly related to success (Garhammer 1993; Kauhanen et al. 2000; McBride et al. 1999; Thomas et al. 1996).

It can be argued that maximum strength is the basic quality that affects power output. Furthermore, maximum strength appears to affect power in a hierarchical manner with diminishing influence as the external load decreases to a point at which other factors such as rate of force development may become more important (Schmidtbleicher 1985, 1992). Therefore, it might be expected that maximum strength would have a greater effect in sports in which relatively large loads must be overcome (i.e. throwing events, American football). However, the exact association(s) between measures of maximum strength and performance are not well understood.

Several categories of approach have been used to investigate these associations. These investigations include 'rank order studies' and correlation studies.

### Rank order studies

One way to begin understanding the potential relationships between strength and sports performance is by descriptive (cross-sectional) studies. If greater maximum strength (or power) makes a difference then strong and powerful teams or athletes will perform better than those teams or athletes that are not as strong or powerful. Although this method does not provide conclusive evidence that a cause

and effect is in operation, we suggest that cause and effect is certainly possible and observation of rank order can affect the construction of a viable hypothesis. For example:

*Example 1.* American collegiate football has three divisions (I, II, III). Division I is made up of the larger universities which grant the most football scholarships, Division II grants fewer scholarships and Division III the least number. Generally, as groups of teams, there are few differences in the type of plays used (strategy) from one division to another. However, if these teams were to play each other on a regular basis then most of the time Division I teams would beat Division II teams which, in turn, would beat Division III teams. If strength (and power) plays a role in winning and losing then a continuum of maximum strength measures should be observed such that Division 1 > Division II > Division III. Fry and Kraemer (1991) studied several hundred American football players including both offensive and defensive positions. Measures of maximum strength- and power-related variables clearly followed the expected continuum (Table 1). It should be noted from Table 1 that the stronger players also had better vertical jump heights and sprint times suggesting a possible relationship between maximum strength and power-speed-related measures.

*Example 2.* It follows that if teams' performances are affected by strength levels then performances of players within a team should also be affected by levels of strength. So, 'first string' players should be stronger and more powerful than 'second string' and so on, again suggesting a continuum of strength (and power) within a football team. Barker et al. (1993) studied a Division IAA university team and divided the players into starters (first string) and non-starters. They found that starters ( $n = 22$ ) had a higher 1 RM squat ( $174.4 \pm 34.5$  vs.  $156.2 \pm 24.6$  kg) than non-starters ( $n = 37$ ), again suggesting that maximum strength plays a role in superior football performance in agreement with Fry and Kraemer (1991). Some evidence indicates that superior strength, especially in relation to body mass, may enhance the ability to perform other motor skills such as jumping (Fry et al. 1991; Stone et al. 1980). Based on the 1 RM squat, normalised by body mass, Barker et al. (1993) also statistically divided the

**Table 1** Performance characteristics of American Football Players (mean ± s.d.) (Fry & Kraemer 1991)

| Test       | Mean                      | Div I                     | Div II                    | Div III                   |
|------------|---------------------------|---------------------------|---------------------------|---------------------------|
| BP (kg)    | 136.9 ± 25.8<br>(n = 776) | 144.5 ± 26.1<br>(n = 283) | 135.2 ± 25.5<br>(n = 296) | 128.6 ± 23.2<br>(n = 197) |
| SQ (kg)    | 185.2 ± 35.7<br>(n = 297) | 192.8 ± 37.6<br>(n = 115) | 182.5 ± 34.4<br>(n = 114) | 176.9 ± 32.4<br>(n = 68)  |
| PC (kg)    | 118.1 ± 17.7<br>(n = 439) | 123.0 ± 17.9<br>(n = 166) | 116.5 ± 17.3<br>(n = 164) | 113.0 ± 16.5<br>(n = 109) |
| VJ (cm)    | 70.2 ± 9.1<br>(n = 505)   | 72.8 ± 9.3<br>(n = 193)   | 69.3 ± 8.5<br>(n = 181)   | 67.4 ± 8.8<br>(n = 131)   |
| 36.6 m (s) | 4.92 ± 0.27<br>(n = 768)  | 4.88 ± 0.27<br>(n = 281)  | 4.92 ± 0.26<br>(n = 282)  | 4.96 ± 0.27<br>(n = 205)  |

Data modified from Fry and Kraemer (1991).  
BP – bench press, SQ – parallel squats, PC – power clean, VJ – vertical jump, 36.6 m – 36.6 m (40 YD) sprint.

team into three relative strength group levels: high, moderate and low (Table 2). Again a continuum is evident as stronger players also had higher vertical jumps compared to moderate- and low-level strength groups.

*Example 3.* For many years, throwers (athletics field events) have been encouraged to participate in strength training programmes in order to enhance throwing ability. Many coaches and athletes strongly believe that increased strength (in specific exercises) is linked to throwing ability. Paul Ward (former Elite Throws Co-ordinator for USA Track and Field) presented evidence in support of this belief that indicated better throwers were stronger (Ward 1982). Ward compiled data from 1978 to 1981, dealing with world and Olympic calibre throwers, which indicated that throwing ability was related to maximum strength in the power clean, snatch, squat and bench press. More recently compiled data (Stone and Stone 1999) supports Ward’s thesis. These data (Table 3, panels a and b) were collected by carefully interviewing (and observing when possible) men and women throwers and their coaches with regard to their

lifting ability (1 RM capability) during 1997–1998. Data in Table 3 deals with throwers in the U.S., which compares different levels of shot-putters and discus throwers. The data shown in Table 3 again indicates that maximum strength may be related to athletic performance.

Correlational studies

A correlation is the strength of the relationship among variables – the correlation coefficient (symbolized as *r*) ranges from –1.0 to 1.0; the closer the coefficient is to 1.0, the stronger the relationship. A positive correlation between two variables would mean they increase together, a negative correlation would mean an inverse relationship. Hopkins (1997) has ranked correlations as *r*:

|                |     |
|----------------|-----|
| Trivial        | 0.0 |
| Small          | 0.1 |
| Moderate       | 0.3 |
| Strong         | 0.5 |
| Very strong    | 0.7 |
| Nearly Perfect | 0.9 |
| Perfect        | 1.0 |

**Table 2** Group performance measures by relative strength (mean ± s.d.)

| Test        | HRS<br>(n = 17) | MRS<br>(n = 27) | LRS<br>(n = 15) |
|-------------|-----------------|-----------------|-----------------|
| SQ (kg)     | 180.9 ± 30.2    | 159.8 ± 27.8    | 148.3 ± 23.4    |
| RS (kg/BdM) | 2.0 ± 0.2       | 1.7 ± 0.1       | 1.4 ± 0.2       |
| VJ (cm)     | 65.8 ± 7.6      | 61.3 ± 7.0      | 55.2 ± 6.8      |
| SVJ (cm)    | 63.8 ± 6.7      | 57.5 ± 6.7      | 52.3 ± 7.0      |

Data modified from Barker et al. (1993).  
SQ – 1 RM parallel squat, RS – relative strength (1 RM SQ/body mass), VJ – vertical jump, SVJ – static vertical jump (3 s pause at 90° knee angle), HRS – high relative strength, MRS – moderate relative strength, LRS – low relative strength.

Table 3 Strength levels of throwers

|                                          | Squat                   | Clean                   | Snatch                  | Bench                   |
|------------------------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
| (a) Shot men (M ± s.d.; kg; 1997–1998)   |                         |                         |                         |                         |
| Godina                                   | 287\327*                | 190                     | –                       | 236                     |
| NLA                                      | 290.3 ± 38.8<br>(n = 3) | 186.0 ± 12.0<br>(n = 3) | 129.3 ± 28.9<br>(n = 2) | 226.8 ± 0<br>(n = 3)    |
| NLP                                      | 283.5 ± 11.3<br>(n = 3) | 155.3 ± 1.8<br>(n = 3)  | 106.8 ± 9.3<br>(n = 2)  | 189.0 ± 15.9<br>(n = 3) |
| Collegiate                               | 266.0 ± 38.4<br>(n = 7) | 137.7 ± 17.3<br>(n = 7) | 84.9 ± 39.3<br>(n = 6)  | 180.8 ± 23.9<br>(n = 7) |
| (b) Shot women (M ± s.d.; kg; 1997–1998) |                         |                         |                         |                         |
| NLA                                      | 168.8 ± 11.7<br>(n = 7) | 106.5 ± 6.7<br>(n = 7)  | 76.5 ± 7.1<br>(n = 7)   | 112.8 ± 9.6<br>(n = 7)  |
| NLP                                      | 147.0 ± 12.3<br>(n = 2) | 100.0 ± 5.8<br>(n = 2)  | 71.1 ± 5.3<br>(n = 2)   | 101.5 ± 5.5<br>(n = 2)  |
| Collegiate                               | 84.5 ± 10.0<br>(n = 5)  | 61.4 ± 4.3<br>(n = 5)   | 46.3 ± 5.9<br>(n = 5)   | 79.8 ± 0<br>(n = 1)     |

Data collected from throwers at UCLA, USC, Wyoming, Appalachian State University. Godina – John Godina – world leader in shot and discus at time of data collection, 327 kg performed with knee wraps; NLA – National Level Automatic Qualifiers – those throwers reaching a previously determined automatic qualifying distance for the USA National Championships, NLP – National Level Provisional Qualifiers – those throwers making a previously determined distance (not automatic) which places them on a list to be considered for the National Championships, Collegiate – good collegiate level throwers not making NLP.

By multiplying the correlation coefficient by itself ( $r^2$ ) the shared variance can be determined. The shared variance is an estimation of how much one variable is explained by another.

Correlational studies can be divided into three categories based on the degree of mechanical specificity used in testing force production and power output: (1) studies in which peak force was measured isometrically and then related to peak force, PRFD or power when measured dynamically within the same exercise context; (2) studies which use the same exercise but in which tests of power or PRFD and the 1 RM were performed at different times; (3) those studies in which strength is measured in one movement pattern (i.e. exercise) and then related to power production, PRFD or performance (i.e. speed, height, distance) in another exercise. Examples of all three types of studies can be considered:

Category 1

A review of the literature generally indicates that isometric measures of maximum strength have only weak to moderately strong correlations with dynamic exercise variables (Wilson and Murphy 1996). However, they point out that isometric–dynamic relationships

can be strengthened by using a test in a position specific (positional specificity) to the performance of interest and by choosing joint angles that involve the highest force outputs in the performance to be used. This would entail isometrically measuring a specific position in the range of motion of the exercise of interest. An example of the use of positional specificity can be found in the paper by Haff et al. (1997) who studied the relationship between peak forces and PRFD using eight well-trained male subjects. In this study (Haff et al. 1997) mid-thigh pulls were performed starting from a knee angle of approximately 144° and a hip angle of approximately 165°. These angles were chosen because of their correspondence to that portion of a clean pull in which the highest forces and RFD are produced. Vertical forces were measured by using a force plate. Force characteristics of the pull were measured isometrically and at 100, 90 and 80% (DP100, DP90, DP80) of the subjects’ best power clean. Isometric peak force (IPF) showed strong correlations with dynamic peak forces generated during DP100, DP90, DP80 ( $r = 0.8, 0.77, 0.66$ , respectively). Additionally, IPF was moderately correlated with dynamic peak PRFD ( $r = 0.36, 0.30, 0.45$ ). Isometric PRFD also showed strong correlations with dynamic peak force ( $r = 0.75, 0.73, 0.65$ , respectively) and was

strongly correlated with dynamic PRFD ( $r = 0.84, 0.88, 0.84$ , respectively). This study indicated that (1) isometric and dynamic peak forces can share some structural and functional foundation and (2) peak force can be related to the ability to produce a high PRFD. In other words, stronger people tend to generate forces faster, a conclusion shared by other researchers (Aagaard et al. 1994).

### Category 2

Moss et al. (1997) investigated the relationship between the 1RM and peak power at various percentages of the 1 RM in elbow flexion. They found very strong correlations between the 1 RM and maximum peak power output ( $r = 0.93$ ). However, they also showed a strong correlation between the peak power output at 2.5 kg and the 1 RM ( $r = 0.73$ ). This latter finding is quite important as it indicated that even at relatively light weights maximum strength (as measured by the 1 RM) has considerable influence on power production.

More recently, Cronin et al. (2000) investigated the role of the 1 RM on the power output during the first 200 ms of a bench press for both plyometric and concentric only conditions. Effects were established for loads representing 40, 60 and 80% of the 1 RM. The results of the study confirmed the enhancement of the concentric phase of the bench press by prior eccentric muscle action (i.e. stretch-shortening). It was also determined that having a high 1 RM augmented power production during the first 200 ms of the concentric phase during a normal (plyometric) bench press. It was concluded that 'for stretch-shortening activity of short duration, greater maximal strength will result in greater instantaneous power production; maximal strength training methods should therefore be an integral training strategy for such activity' (p. 1769).

### Category 3

Considering the strong theoretical underpinning and experimental data (categories 1–2) it is logical to assume that maximum strength contributes markedly to strength/power sports performances. However, experimental evidence in which maximum

strength or estimates of maximum strength (i.e. 1 RM) have been related to performance or with other performance-related variables is difficult to find, especially studies using well-trained athletes. Several available studies have focused on the relationship of maximum strength and jumping. Seyforth et al. (2000), studying the long jump and using mathematical modelling techniques have provided a strong theoretical basis, which indicates that maximum strength is a primary factor in jumping performance. They found that maximum strength, particularly eccentric strength, was more important than factors such as tendon compliance or muscle contraction speed in improving long jump performance.

Although not all studies agree (Costill et al. 1968; Hutto 1938; Start 1966), several investigations (Berger & Blaschke 1967; Berger & Henderson 1966; McClements 1966; Thomas et al. 1996) using the unweighted standing long jump and vertical jump indicated a strong relationship ( $r = 0.7$ ) between power and measures of maximum strength. Whitely and Smith (1966) and Eckert (1968) found that by adding additional resistance to a movement, the relationship between maximum strength and power, and strength and speed tended to increase with the added resistance, a finding supported by Schmidtbleicher (1985, 1992). However, these studies used untrained subjects and measured maximum strength in various ways. Stone et al. (1998 – unpublished data) investigated the relationship of the 1 RM squat and the standing long jump (SLJ) among trained (college sprinters,  $n = 12$ ) and relatively untrained men and women (4 weeks in a beginning weight training class,  $n = 21$ ). The correlation between the 1 RM squat and SLJ was  $r = 0.66$  for the weight training class,  $r = 0.72$  for the sprinters and  $r = 0.82$  for the combined groups ( $n = 33$ ). Thus, there is evidence that during jumping activities, 50% or more (i.e. shared variance) of the performance is due to maximum strength and this relationship can increase with the load.

More recently, Glaister et al. (2000 – unpublished data) studied the relationship between the 1 RM parallel squat and tests of agility and jumping capabilities, using the Scottish National Badminton team ( $n = 13$ ). This study was part of the ongoing sports



testing/sports science programme initiated by the Scottish Institute of Sport. The results (Glaister et al. 2000) indicated that the 1 RM squat was strongly related to weighted and unweighted counter movements and static vertical jumps as well as tests of agility ( $r = 0.65 - 0.87$ ). Furthermore, the 1 RM was strongly related to the ability to repeat multiple agility runs. These strong correlations were found both for absolute and relative (1 RM/kg body mass) values and jumping and agility variables (Table 4). Glaister et al.'s results indicate that 1 RM squat strength and 1 RM squat/kg body mass have significant relationships with power, speed and speed-endurance related variables. These data suggest that maximum leg and hip strength (1 RM squat) may be a significant component in jumping height and agility capabilities. Therefore, development of the 1 RM squat may produce improved capabilities for jumping and agility performance.

The relationship between sprinting and measures of maximum strength has also been studied. A theoretical foundation supporting a strong relationship between strength and performance can be found in the work of Weyand et al. (2000). Using a mathematical model as well as experimental evidence from a treadmill mounted force plate, Weyand et al. (2000) found that peak ground reaction forces (vertical forces affecting flight time and stride length) were the limiting factors in running speed. The peak ground reaction forces were influenced by the maximum available force (maximum force which can be produced) and the rate of force development. Since dynamic peak force and PRFD can be strongly related to measures of maximum strength (Haff et al. 1997), the running speed may be associated

with maximum strength. Additionally, investigations of the relationship between 'explosive strength' (various types of weighted and unweighted jumps) and sprinting ability have shown strong to very strong correlations (Baker & Nance 1999; Manou et al. 2000). Because maximum strength and jumping ability have strong correlations (Stone et al. 1998) it is logical to assume that maximum strength should be related to sprinting ability.

Several investigators have studied the relationship between maximum strength measured using isokinetic devices and sprint performance. Farrar and Thorland (1987) found that, in active but non-sprint trained subjects, a poor relationship between peak leg extension torque and 100 m times at fast speeds (5.24 rad/s) or slow speeds (1.05 rad/s). However, the faster sprinters did show peak torques at the slow leg extension speed that were significantly greater than the slower sprinters.

Delecluse (1997), citing unpublished data on physical education students, studied the relationship of concentric isokinetic knee and ankle extension (5.24 and 3.49 rad/s), knee flexors (1.13 rad/s) and running speed over 40 m. The data indicated that initial acceleration (first 15 m) was related to knee and ankle extensor strength, and that flexor strength was related to the speed during the final 20 m. Dowson et al. (1998) studied a heterogeneous group of athletes consisting of rugby players, sprinters and physically active young men. They found that performance for 0–15 m and velocity over 30–35 m were significantly related ( $r = -0.41$  to  $-0.69$ ) to absolute and relative strength (torque/body mass) of several movements. Furthermore, it was shown that the strength of these relationships could be

**Table 4** Relationship of 1 RM squat measures to jumping and agility measures (Glaister et al. 2000)

|         | CMJ  | WCMJ | CMJP | WCMJP | SJ   | WSJ  | SJP  | WSJP | T test | X test | RX test |
|---------|------|------|------|-------|------|------|------|------|--------|--------|---------|
| 1 RM    | 0.73 | 0.80 | 0.86 | 0.86  | 0.71 | 0.76 | 0.86 | 0.87 | -0.72  | -0.79  | -0.69   |
| 1 RM/kg | 0.74 | 0.75 | 0.66 | 0.66  | 0.71 | 0.74 | 0.67 | 0.67 | -0.75  | -0.85  | -0.65   |

1 RM – 1 repetition maximum squat, 1 RM/kg – 1 RM divided by body mass, CMJ – counter movement vertical jump, WCMJ – weighted (10 kg) countermovement jump, CMJP – countermovement jump power, WCMJP – weighted (10 kg) countermovement jump power, SJ – static vertical jump, WSJ – weighted (10 kg) countermovement jump, SJP – countermovement jump power, WSJP – weighted (10 kg) countermovement jump power, T-test – agility, X-test – badminton specific agility, RX-test – repeated X test (repeating the test 15 times with a 14 s interval for males and a 16 s interval for females).

**Table 5** Relationship between estimates of maximum strength and sprint times (Baker & Nance 1999)

| Strength measure | 10 m  | 40 m  |
|------------------|-------|-------|
| 3 RM squat       | −0.06 | −0.19 |
| 3 RM squat/BdM   | −0.39 | −0.66 |
| 3 RM HC          | −0.36 | −0.24 |
| 3 RM HC/BdM      | −0.56 | −0.72 |

BdM – body mass, HC – hang clean.

improved by using an allometric force model, which considered differences in limb length and body mass. These movements included both concentric and eccentric knee flexion and extension torque measured at a variety of speeds ranging from 1.05 rad/s to 4.19 rad/s. Similar findings have been reported by Alexander (1989) using ‘elite’ male (10.83 s for 100 m) and female (12.03 s for 100 m) sprinters.

Although these data indicate that peak torque can have moderate to strong correlations with sprint performance, the use of isokinetic dynamometers for strength testing must be questioned, particularly in trying to relate peak isokinetic torques to sports performance (Stone et al. 2000). For example, typical isokinetic testing procedures use single joint, open kinetic chain movements. However, sprinting or jumping are multi-joint activities with propulsive phases, which are largely closed-chain activities. Furthermore, most of these isokinetic studies did not use strength measures in which forces were applied vertically. In addition, the contribution of eccentric strength is seldom considered.

One might argue that because vertical forces have been shown to be limiting factors in sprinting, there should be a relationship between measures of maximum ‘vertical strength’ and sprint performance.

Using trained Australian Rugby players ( $n = 20$ ), Baker and Nance (1999) found only weak correlations between absolute estimates of maximum strength (3 RM squat and hang clean), and sprint times over 10 and 40 m. However, when strength measures were normalised by body mass stronger correlations were noted (Table 5). This study points out two interesting possibilities:

1. Sprint performance may be more related to relative (‘normalized’) measures of

maximum strength. In this context it can be argued that simply dividing by body mass does not necessarily obviate regional body mass differences (for example, some people have relatively more mass and lean body mass in the upper or lower body). Nor does maximum strength increase in a linear fashion with body mass. Thus, other methods of accounting for differences in body mass may be necessary (Dowson et al. 1998).

2. The hang clean was better correlated to sprint performance than the squat. However, weightlifting movements (snatch, clean and jerk) and their variations such as hang cleans may be more accurately described as ‘Explosive Strength’ or high power exercises. In this context Baker and Nance (1999) also found that the power output/kg generated during weighted jumps (40–100 kg) had correlations with the 10 m sprint ranging from  $r = -0.52$  to  $-0.61$  and  $r = -0.52$ – $-0.75$  for the 40 m sprint.

Longitudinal studies

Correlations only indicate a magnitude of relationship and do not necessarily indicate a cause and effect. In order to better understand ‘cause and effect’ longitudinal studies are necessary. It is not the purpose of this paper to provide a substantial review of the many longitudinal studies dealing with increased strength and its effects on other performance variables. As with cross-sectional studies many factors can affect the outcome. These factors include trained vs. untrained subjects, length of study, and the degree of mechanical specificity of the exercises used in training and testing. It should also be noted that in no study has strength training altered selected performance variables (i.e. sprinting, jumping, agility) to the same extent as the changes observed in maximum strength (i.e. the changes are not perfectly correlated). This indicates that changes in other factors (i.e. power, PRFD) may also accompany the increases in strength resulting from training that also contribute to improved performance. It is also possible that the lack of direct correspondence between increased strength and other types of performance is at

least partially due to a lag time (Abernethy & Jurimae 1996; Delecluse 1997; Sanborn et al. 2000; Stone et al. 2000). Lag time is concerned with a period of time in which an athlete 'learns' how to use increased strength in various sports events. It is possible that this lag time may extend many months; if this is true then this would be beyond the limited experimental bounds of most studies which typically only last a few weeks.

Several studies have examined the effects of resistance training on a number of different performance variables such as jumping, test of speed, power and agility, generally these studies have shown that an increase in strength is accompanied by an increase in performance among relatively untrained subjects (for example see Augustsson et al. 1998; Robinson et al. 1995; Sanborn et al. 2000; Stone et al. 1980). Making changes in advanced or elite athletes is more difficult (Baker 1996) and requires more creative training programmes. However, it appears that a key ingredient in these advanced programmes is improvement of maximum strength as well as specialised work on speed and power (Harris et al. 2000).

This review has considered evidence from a number of different types of research as well as observational data. Collectively, the data indicate that maximum strength can be strongly related to sports performances that rely on speed and power. Although explaining performance in strength/power sports is a multi-factorial problem there is little doubt that maximum strength is a key component.

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